



AI BASED REAL TIME VEHICLE DETECTION AND COUNTING FOR SMART TRAFFIC MONITORING USING OPEN CV

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Agenda

Introduction

- Importance of Computer Vision in Traffic Management

Literature Review

- Existing Traffic Monitoring Systems
- AI -Based Approaches
- Research Gaps

Problem Statement

- Challenges in Conventional Traffic Monitoring

Objectives

- Primary Objective
- Specific Objectives

System Architecture

- Overall Project Workflow
- Architecture Diagram

Modules Overview

- **Module 1:** Video Acquisition and Image Preprocessing
- **Module 2:** AI-Based Vehicle Detection
- **Module 3:** Vehicle Tracking and Counting
- **Module 4:** Traffic Analysis and Monitoring
- **Module 5:** Result Visualization and Performance Evaluation

Introduction

- Rapid urbanization has significantly increased the number of vehicles on roads.
- Traditional traffic monitoring systems rely on manual observation or sensor-based infrastructure, which can be expensive and difficult to maintain.
- Artificial Intelligence (AI) and Computer Vision enable automatic vehicle detection, tracking, and counting using video feeds.
- OpenCV provides powerful image processing capabilities for real-time traffic monitoring.
- This project develops an intelligent traffic monitoring system that detects and counts vehicles automatically from surveillance videos.

Literature Review

Existing Systems

- Manual vehicle counting by traffic personnel.
- Sensor-based traffic monitoring systems.

AI-Based Approaches

- Faster R-CNN
- SSD (Single Shot Detector)
- OpenCV Deep Neural Network (DNN)

Research Gap

- High implementation cost.
- Limited real-time performance and reduced accuracy in complex traffic conditions.
- Need for a low-cost and efficient AI-based monitoring system.

Problem Statement

- Manual traffic monitoring is inefficient and time-consuming.
- Existing systems require costly hardware and infrastructure.
- Traditional techniques struggle with varying lighting and weather conditions.
- Real-time traffic statistics are difficult to obtain.
- There is a need for an intelligent, automated vehicle detection and counting system for efficient traffic management.

Objectives

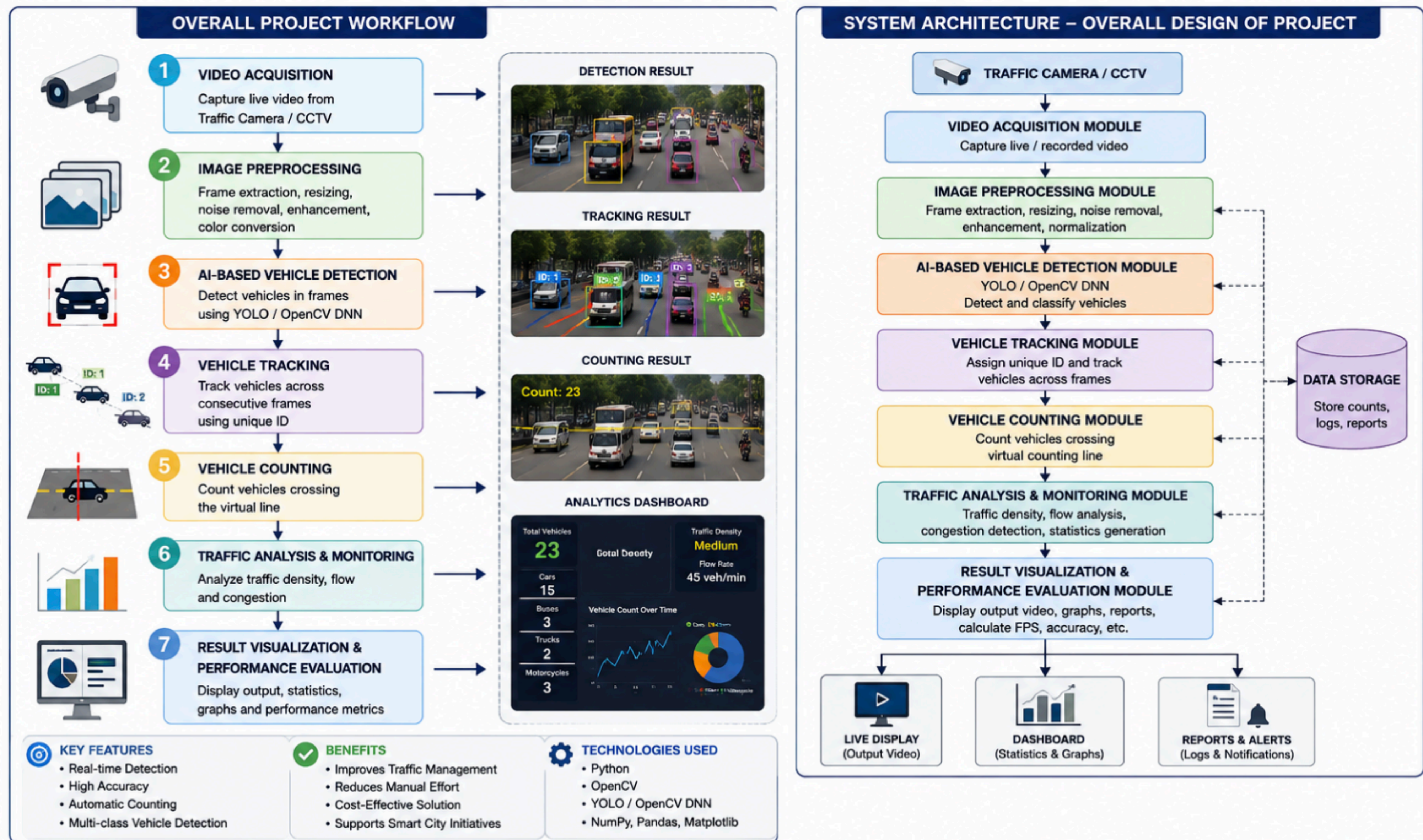
Primary Objective

To develop an AI-based real-time vehicle detection and counting system using OpenCV for smart traffic monitoring.

Specific Objectives

- Capture traffic video.
- Preprocess video frames.
- Detect vehicles automatically.
- Track vehicle movement.
- Count vehicles accurately.
- Analyze traffic density.
- Display real-time traffic statistics.
- Evaluate system performance.

System Architecture - Overall Design of Project



Modules Overview

- **Module 1:** Captures traffic videos and preprocesses frames by resizing, enhancing image quality, and removing noise.
- **Module 2:** Detects and classifies vehicles such as cars, buses, trucks, and motorcycles using AI-based object detection models.
- **Module 3:** Tracks detected vehicles across video frames and accurately counts them as they cross a virtual counting line.
- **Module 4:** Analyses vehicle count, traffic density, and traffic flow to monitor road conditions and identify congestion.
- **Module 5:** Displays the processed results through annotated videos, graphs, dashboards, and evaluates the system using performance metrics like accuracy, precision, recall, F1-score, FPS, and processing time.